



Fake video from image

Researchers have created an AI that can make realistic fake videos based on a single image. <https://digit.in/june19-60>



Brain Implants

In as little as 20 years, brain implants could make learning, and search engines obsolete. <https://digit.in/june19-61>

6. Azure Emotion API: This API takes the input as a facial expression in an image and returns a set of emotions for each face in the image along with its confidence value. It also returns a bounding box for the face, using the Face API.

7. Azure Video Indexer: It provides extraction of following insights from videos. It is a cloud service.

8. Google Cloud Vision API: It encapsulates various powerful machine learning models in a simple REST API and enabling one to understand the content of an image. This API also classifies images into a large number of categories like sailboat, lion, Eiffel Tower, and many more. It even detects individual objects and faces that are embedded within images and reads printed words which are contained inside images. This has many applications such as helping users in building metadata on image catalogues, moderating offensive content on various social media websites, and enabling new marketing scenarios with the help of image sentiment analysis.

9. Google Cloud Video Intelligence: It enables one to search and discover videos by extracting metadata of videos with a simple REST API. Some of the other MLaaS microservices offered by Google, Microsoft, Amazon and IBM are as follows:

1. Amazon Machine Learning:

Its main aim is to provide all developers irrespective of their skill levels access to use machine learning technology. This service provides visualization tools and wizards that aid in the process of creating machine learning models without the need to learn complex Machine Learning technology and algorithms.

2. Azure Batch AI: This managed AI service enables data scientists and researchers to train Artificial Intelligence and other machine learning models that require high computing capacity on clusters of Azure virtual machines, including

Virtual machines with GPU support. One only needs to specify the location of the inputs and the outputs, along with the requirements of the job. The Batch AI handles the rest of the things.

3. Google Cloud Job Discovery: Its main aim is to help people find jobs easily through its platform. It provides for a plug and plays access to Google's search and machine learning capabilities. Hence, the entire talent discovery ecosystem is enabled to tracking applicants and staffing agencies along with company career sites and portals can enhance engagement and conversion rate of candidates.

4. IBM Watson Studio: It can be used by data scientists to scale to thousands of training iterations. It also contains a Deep Learning Assistant that manages and assists throughout the experiment in Watson Studio. It also performs tasks like:

1. Distribution of source code and data to GPU processors with selected NVIDIA or Tesla GPU.
2. Automatically shuts down the machine instance when it is no longer required to save money bills.
3. Keeps track of all the hyperparameters associated with each training iteration.
4. Extracts events from training log and graphically show them in Watson Studio.

MLaaS will be instrumental in Artificial Intelligence adoption in the near future since it makes it easier for businesses and developers to take leverage the power of machine learning capabilities without requiring any machine learning expertise or even programming experience in some cases. It will allow organizations to use data in new ways, using machine learning that would have been impossible in the past without a skilled AI developer.

With the microservices of machine learning services platforms like Microsoft Azure, Google Cloud Platform, Amazon Web Services and IBM Wat-

son, it has become easy to set up and run machine learning algorithms.

This has translated to enhanced business processes and improved customer interaction. Aiding in business strategy without additional support on high computational power involvement, MLaaS will become instrumental in the near future, and we will see tech companies investing heavily in developing better Machine Learning as Service platforms and tools.

CONCLUSION

All the above platforms provide machine-learning services that enable users to leverage the power of machine learning in their applications without the need of any programming skills or data science expertise. For the developing of machine learning models, Azure Machine Learning Studio has an advantage over others due to its unique feature like Cortana Intelligence Suite that helps in doing things faster and in a simpler way, its drag and drop components and good processing speed. However, among all the service providers, Google Cloud Machine Learning Engine provides the fastest processing speed since it uses specially designed TPU – Tensorflow Processing Unit for AI tasks. For accomplishing Natural Language Processing tasks, Microsoft Azure provides many different APIs with each for a specific work, like translation, analysis, analytics, etc.

For translation purposes, Google Cloud Translation API can be considered because of its high responsiveness and many languages support. For Computer Vision tasks, Google Cloud Vision API can classify images into the highest number of categories among others. Amazon Rekognition is great for image and video analysis. IBM Watson Studio is the preferred choice for deep learning applications because of its Deep Learning Assistant that accomplishes a variety of tasks as listed above and it is highly scalable. Overall, all the platforms are equally good for beginner and intermediate level services. 